

Soudeep Deb

CURRENT POSITION	Associate Professor and Chairperson Decision Sciences Area Indian Institute of Management Bangalore, India.	Webpage: soudeepd.github.io Phone: +91 80-26993387 E-mail: soudeep@iimb.ac.in
RESEARCH INTERESTS	Time series, Spatio-temporal models, Nonparametric methods, Statistical machine learning, Sports analytics, Applications of statistics in real estate, finance, environmental science, and other fields.	
EXPERIENCE	Indian Institute of Management Bangalore , Bengaluru, KA, India.	Mar 2020 - Present
	- Chairperson, Decision Sciences Area. - Associate Professor, Decision Sciences Area. - Assistant Professor, Decision Sciences Area.	Apr 2024 - Present Jul 2024 - Present Mar 2020 - Jul 2024
	NBC Universal Media, LLC. , New York, NY, USA.	Sep 2018 - Feb 2020
	Senior Lead Data Scientist, Decision Sciences Division.	
EDUCATION	University of Chicago , Chicago, IL, USA.	
	Ph.D., Statistics	Aug 2018
	- Thesis: Irregular spaced random field, Spatio-temporal data and Clustering of time series - Advisory committee: Dr. Wei Biao Wu, Dr. Ruey S. Tsay, Dr. Michael L. Stein	
	Indian Statistical Institute , Kolkata, WB, India.	
	Master of Statistics (M. Stat.)	May 2013
	- First Division with Distinction, Specialization: Mathematical Statistics and Probability - Thesis: Association analysis for identifying rare genetic variants, Advisor: Dr. Saurabh Ghosh	
	Bachelor of Statistics (B. Stat.)	May 2011
	First Division with Distinction	
BOOKS	(With S. Das) Business Analytics: Data to Decisions. First edition in 2025, published by Universities Press (India) Private Limited. ISBN: 9789393330994. [Amazon link] (With D. Dey) Mathematical Techniques for Competitive Examinations. First edition in 2023, published by Orient Blackswan Pvt Ltd. ISBN: 9393330107. [Amazon link]	
RESEARCH GRANTS	India-Israel Joint Research Cooperation (IJRC) Grant , INR 1.6 million.	2024 – 2026
	- From Department of Science & Technology, India, and Ministry of Innovation, Science & Technology, Israel. - Co-PI: Prof. Itai Dattner, University of Haifa, Israel. - Project: Hybrid-AI Quantile Regression Combining Nonparametric Statistical Methods and Physics-Informed Neural Networks for Analysing Spatio-temporal Climate Data.	
	Research Challenge Grant from IIM Bangalore, India, INR 1.5 million.	2025 – 2026
	- Co-PI: Prof. Anand Deo, IIM Bangalore, India. - Project: Estimating spatial extremes using EVT-based scaling.	
	Research Travel Grant from London Mathematical Society, UK, GBP 2200.	2024
	Offered under Scheme 5 ‘Collaborations with Developing Countries’ Grant.	
	Research Seed Grant from IIM Bangalore, India, INR 0.5 million.	2024 – 2025
	Project: Nonparametric methods of structural break detection in time series.	
	Research Seed Grant from IIM Bangalore, India, INR 0.3 million.	2021 – 2023
	Project: New techniques to analyze categorical and discrete time series data.	
	Young Faculty Research Chair Grant from IIM Bangalore, India, INR 0.9 million.	2020 – 2023

STUDENTS

PhD thesis advisor/co-advisor:

- Anchal Soni, Decision Sciences, Indian Institute of Management Bangalore. Feb 2023
Thesis: Methods of analyzing structural breaks in multivariate time series: Applications to financial data.
- Siddharth Rawat, Decision Sciences, Indian Institute of Management Bangalore. May 2023
Thesis: Spatio-temporal models in epidemiology and climate change.
- Kapil Gupta, Decision Sciences, Indian Institute of Management Bangalore. Mar 2025
Thesis: Analyzing house price dynamics using novel spatio-temporal methods.
- Archi Roy, Dept of Mathematics, Indian Institute of Science Education & Research Pune. Aug 2025
Thesis: Nonparametric methods for short and long-range dependent time series: Applications in financial and biomedical datasets.
- Kunal Rai, Decision Sciences, Indian Institute of Management Bangalore. (Exp) 2026
Thesis: Nonparametric Quantile Regression in Time Series: Theory and Applications.
- Chinmay Divekar, Decision Sciences, Indian Institute of Management Bangalore. (Exp) 2026
Thesis: Novel methods for within-game analytics problems in sports.
- Amrutha Seshagiri, Decision Sciences, Indian Institute of Management Bangalore. (Exp) 2028
Thesis: Essays on tourism analytics.
- Thaha Uwaise, School of Advanced Sciences, Vellore Institute of Technology. (Exp) 2029
Thesis: New machine learning methods in analyzing sports data.

Doctoral committee member:

- M Mareeswaran, Finance & Accounting, Indian Institute of Management Bangalore. Mar 2025
- Sajad S Santhosh, Public Policy, Indian Institute of Management Bangalore. Jul 2025
- Sabhya Rai, Economics, Indian Institute of Management Bangalore. Mar 2026
- Kalaga Venkata Laxmana Tejaswi, Marketing, Indian Institute of Management Bangalore. (Exp) 2028
- Mantu Prasad Gupta, Decision Sciences, Indian Institute of Management Bangalore. (Exp) 2028

MS thesis advisor:

- Snehil Maurya, Indian Institute of Science Education & Research Mohali. (Exp) 2026
- Ashish Kumar Yadav, Indian Institute of Science Education & Research Mohali. (Exp) 2027

PAPERS UNDER REVIEW

- Roy, S., **Deb, S.**, Karmakar, S., Roy, R. (2026). Nonparametric regression of spatio-temporal data using infinite-dimensional covariates. [\[preprint\]](#)
- Basu, T., **Deb, S.**, Roy, R. (2026). Optimising football transfer strategy under budget constraints: A weighted multi-criteria approach. [\[preprint\]](#)
- Rai, K., Roy, A., Dattner, I., **Deb, S.** (2026). A nonparametric approach to understand multivariate quantile dynamics in financial time series. [\[preprint\]](#)
- Rai, K., Anand, A., **Deb, S.** (2026). Multiperiod volatility forecasting with optimization-based model selection: Evidence from NIFTY-50 Banks. [\[preprint\]](#)
- Gupta, K., Srivastava, S., **Deb, S.**, Panchapagesan, V. (2025). CPRI-Office: A new commercial property rental index for Indian cities using spatio-temporal modeling techniques. [\[preprint\]](#)
- **Deb, S.**, Neves, C, Roy, S (2024) Nonparametric quantile regression for spatio-temporal processes. [\[preprint\]](#)
- Roy, A., Podder, M., **Deb, S.** (2024). Nonparametric method of structural break detection in stochastic time series regression model. [\[preprint\]](#)
- Rawat, S., Durrant, A., Simpson, A., Nielson, G., Berrett, C, **Deb, S** (2023) A Bayesian approach to identify changepoints in spatio-temporal ordered categorical data: An application to COVID-19 data. [\[preprint\]](#)
- Bag, S., Gupta, K., **Deb, S.** (2022). A review and recommendations on variable selection methods in regression models for binary data. [\[preprint\]](#)

JOURNAL & CONFERENCE PUBLICATIONS

- **Deb, S.** (2026). Skill or chance? A Bayesian analysis of dependence and heterogeneity in penalty shootouts in football. Journal of Sports Analytics, 12. [\[publication\]](#)
- Panja, M., Chakraborty, T., Biswas, A., **Deb, S** (2026) E-STGCN: extreme spatio-temporal graph convolutional networks for air quality forecasting. Journal of the Royal Statistical Society Series A, qnag010. [\[publication\]](#)
- **Deb, S.**, Das, S. (2026). Optimal selection of the starting lineup for a football team. IIMB Management Review, 100642. [\[publication\]](#)

- Basu Sarbadhikary, S., Roy, A., **Deb, S.** (2026) A data-driven approach to spatial zoning and anomaly detection in the dynamic real estate network. *Environment and Planning B: Urban Analytics and City Science*, 23998083251411954. [\[publication\]](#)
- Sen, S., **Deb, S.** (2025) tSNE-Spec: A new classification method for multivariate time series data. *Journal of Multivariate Analysis*, 105537. [\[publication\]](#)
- Gupta, K., **Deb, S.** (2025) A divide-and-conquer approach for spatio-temporal analysis of large house price data from Greater London. *Annals of Operations Research*, 1-42. [\[publication\]](#)
- Deshmukh, S. S., **Deb, S.** (2025) A survey of statistical and machine learning methods of quantile regression in time series and their suitability in predicting dengue outbreaks. *Japanese Journal of Statistics and Data Science*, 1-49. [\[publication\]](#)
- **Deb, S.**, Jana, K. (2024) Nonparametric quantile regression for time series with replicated observations and its application to climate data. *Statistical Science*, 39(3), 428-448. [\[publication\]](#)
- **Deb, S.**, Roy, R., Das, S. (2024). Forecasting elections from partial information using a Bayesian model for a multinomial sequence of data. *Journal of Forecasting*, 43(6), 1814-1834. [\[publication\]](#)
- Divekar, C., **Deb, S.**, Roy, R. (2024) Real-time forecasting within soccer matches through a Bayesian lens. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 187(2), 513–540. [\[publication\]](#)
- Chattopadhyay, A., **Deb, S.** (2024) A spatio-temporal model for binary data and its application in analyzing the levels of COVID-19 spread. *ASTA Advances in Statistical Analysis*, 108(4), 823-851. [\[publication\]](#)
- Lakshmi M.V., **Deb, S.**, Sen, R. (2024) Environmentally Responsible Index Tracking: Maintaining Performance while Reducing Carbon Footprint of the Portfolio. *Statistics and Applications* (forthcoming).
- Gupta, K., Krishnamurthy, V., **Deb, S.** (2024) What elements of the opening set influence the outcome of a tennis match? An in-depth analysis of Wimbledon data. *IIMB Management Review*. [\[publication\]](#)
- Roy, A., **Deb, S.**, Chakarwarty, D. (2024) Impact of COVID-19 on public social life and mental health: A statistical study of Google Trends data from the USA. *J. Appl. Stat.*, 51(3), 581-605. [\[publication\]](#)
- Roy, A., Soni, A., **Deb, S.** (2023) A wavelet-based methodology to compare the impact of pandemic versus Russia-Ukraine conflict on crude oil sector and its interconnectedness with other energy and non-energy markets. *Energy Economics*, 124, 106830. [\[publication\]](#)
- **Deb, S.**, Karmakar, S. (2023) A novel spatio-temporal clustering algorithm with applications on COVID-19 data from the United States. *Computational Statistics & Data Analysis*, 107810. [\[publication\]](#)
- **Deb, S.** (2023). Analyzing airlines stock price volatility during COVID-19 pandemic through internet search data. *International Journal of Finance & Economics*, 28(2), 1497-1513. [\[publication\]](#)
- Mareeswaran, M., Sen, S., **Deb, S.** (2023). New methods of structural break detection and an ensemble approach to analyze exchange rate volatility of Indian rupee during COVID-19. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 187(1), 39-61. [\[publication\]](#)
- Rawat, S., **Deb, S.** (2023). A spatio-temporal statistical model to analyze COVID-19 spread in the USA. *Journal of Applied Statistics*, 50(11-12), 2310-2329. [\[publication\]](#)
- **Deb, S.**, Majumdar, M. (2023). A quadratic trend-based time series method to analyze the early incidence pattern of COVID-19. *Biostatistics & Epidemiology*, 7(1), e2076529. [\[publication\]](#)
- Majumdar, M., Banerjee, M., Sengupta, J., **Deb, S.**, Jana, C. K., Roy, B. K. (2023) Prevalence and spectrum of diabetic peripheral neuropathy and its correlation with insulin resistance – An experience from eastern India. *International Journal of Advanced Research*, 11(06), 1085-1094. [\[publication\]](#)
- **Deb, S.**, Deb, S. (2022). An ensemble method for early prediction of dengue outbreak. *Journal of the Royal Statistical Society Series A*, 185(1), 84-101. [\[publication\]](#)
- **Deb, S.** (2022). A goal based index to analyze the competitive balance of a football league. *Journal of Quantitative Analysis in Sports*, 18(3), 171-186. [\[publication\]](#)
- Nahata, S., **Deb, S.** (2021) A Machine Learning Approach to Analyze the Effect of Situational and Player-Dependent Features on Converting Freekicks in Soccer. In *Conference Proceedings 2021 Asia-Singapore Conference on Sport Science* (p. 19). [\[publication\]](#)

- **Deb, S.**, Tsay, R. S. (2019). Spatio-temporal Models with Space-time Interaction and Their Applications to Air Pollution Data. *Statistica Sinica*, 29, 1181-1207. [\[publication\]](#)
- **Deb, S.**, Dey, D. (2019). Spatial Modeling of Shot Conversion in Soccer to Single out Goalscoring Ability. *Journal of Sports Analytics*, 5(4), 281-297. [\[publication\]](#)
- **Deb, S.** (2019). VAR Model Based Clustering Method for Multivariate Time Series Data. *Journal of Mathematical Sciences*, 237(6), 754-765. [\[publication\]](#)
- Prickett, K.C., Guitierrez, C., **Deb, S.** (2019). Family Firearm Ownership and Firearm-related Mortality among Young Children: 1976-2016. *Pediatrics*, 143(2), e20181171. [\[publication\]](#)
- Chazin, H., **Deb, S.**, Falk, J., Srinivasan, A. (2019). New Statistical Approaches to Intra-individual Isotopic Analysis and Modelling of Birth Seasonality in Studies of Herd Animals. *Archaeometry*, 61(2), 478-493. [\[publication\]](#)
- **Deb, S.**, Pourahmadi, M., Wu, W. B. (2017). An Asymptotic Theory for Spectral Analysis of Random Fields. *Electronic Journal of Statistics*, Vol. 11, No. 2, p. 4297-4322. [\[publication\]](#)
- Zechner, C., **Deb, S.**, Koeppl, H. (2013). Marginal Dynamics of Stochastic Biochemical Networks in Random Environments. In *Control Conference (ECC), 2013 European*, p. 4269-4274, IEEE. [\[publication\]](#)

TEACHING

Degree-granting Programmes, at Indian Institute of Management Bangalore:

- Decision Sciences I (Core for MBA students) Fall, 2020 – 2025
- Sports Analytics (Elective for MBA students) Fall, 2023 – 2025
- Advanced Statistics for Business (Core for BBA students) Winter, 2025
- Nonparametric Statistics (Core for PhD students) Spring, 2025
- Multivariate Statistics (Core for PhD students) Spring, 2021 – 2024
- Advanced Statistical Methods & Computing (Core for PhD students) Fall, 2024
- Statistical Inference (Core for PhD students) Winter, 2020

Executive Education Programmes, at Indian Institute of Management Bangalore:

- Data Science & Artificial Intelligence (Co-director) 2025 – 2027
- IIMB-CREDAI Business Leadership Programme (Co-director) 2025 – 2027
- MDP for Food Corporation of India (8 sessions) Apr 2024
- Business Analytics and Intelligence (3 sessions) Fall, 2022 – 2024

Other courses:

- Introductory Statistics (UG), University of Chicago 2015, 2017
- Statistical Models and Methods I (UG), University of Chicago 2015

CONSULTING SERVICES

National Foundation for Corporate Governance, Ministry of Corporate Affairs, India 2025

- Topic: Feedback survey and concurrent evaluation of the Prime Minister Internship Scheme.

Smart City Mission, Ministry of Housing & Urban Affairs, India 2024

- Topic: Improvement in quality of education through smart classrooms.

DBS Bank, Singapore 2023

- Topic: Value of data joint study.

Real Estate Research Initiative (IIMB-RERI), Bengaluru, India Since 2023

- Topic: Developing a commercial rental index through spatial modeling.

SportsKPI, Bengaluru, India 2022-23

- Topic: Valuation of a Kabaddi player and its use in building strategies.

LATEST SEMINARS

- **May 2026.** Optimizing Football Squad Selection: A Multi-Objective Approach to Transfer Strategy under Budget Constraints. AUEB Sports Analytics Workshop & Conference 2026 (SAW 2026), Athens, Greece.
- **Dec 2025.** Bridging search behavior and market dynamics: A hybrid model for high-frequency financial data. CFE-CMStatistics Conference, London, UK.
- **Dec 2025.** Real-time forecasting in football matches. Liverpool Management School, Liverpool, UK.

- **May 2025.** Data Handling, Visualisation, and Storytelling in Academic Research. Workshop-cum-training session at RV University, Bengaluru, India.
- **Dec 2024.** Nonparametric Regression of Spatio-temporal Data using Infinite-dimensional Covariates. IISA Conference 2024, Cochin, India.
- **Dec 2024.** Structural breaks in the spatial network of real estate dynamics: A study of UK property transactions. Essex Data Science Seminar Series, University of Essex, Colchester, UK.
- **Sep 2024.** Nonparametric changepoint detection in time series data. COMPSTAT 2024, Giessen, Germany.
- **Jul 2024.** A workshop on ‘Data summarisation and Visual Storytelling using Statistics’, PES University Research Workshop Series, Bengaluru, India.
- **Jun 2024.** Real-time forecasting in sports. AI Frontiers Series, Commonwealth Bank of Australia (virtual).
- **May 2024.** A workshop on Sports Analytics. Conference on Excellence in Research & Education, Indian Institute of Management Indore, India.

HONORS

At Indian Institute of Management Bangalore:

- Excellence in Teaching Recognition Multiple years since 2022

At University of Chicago:

- International House Ralph W. Nicholas Fellowship 2017-18
- Graduate Council Travel Fund Award 2017
- Runner-up for Department of Statistics Consulting Award 2016
- Nominated for Best Teaching Assistant in Physical Sciences Division Winter 2014

Earlier awards:

- Kishore Vaigyanik Protsahan Yojana scholarship, Indian Institute of Science 2007 to 2013
- Selected for International Mathematical Olympiad Training Camp, India 2007 & 2008

PROFESSIONAL SERVICES

Administrative roles, at Indian Institute of Management Bangalore:

- Chairperson, Decision Sciences Area. Apr 2024 - present
- Member, Postgraduate Programme Committee. Apr 2025 - present
- Doctoral Program Coordinator, Decision Sciences Area. 2023-24
- Mentor in Pre-doctoral Programme, Decision Sciences Area. 2021 - 2024

Editorial Board member, for the following journal:

- BMC Public Health, Associate Editor. 2026 – present
- Journal of Sports Analytics, Associate Editor. 2025 – present
- Sankhya B, Associate Editor. 2025 – present
- Nature Scientific Reports, Associate Editor in the Climate Change track. 2024 – present

Session chairs, at the following conferences:

- CFE-CMStatistics Conference, London, UK Since 2021
- IISA Conference, Cochin, India. Dec 2024
- EcoSta 2024, Beijing, China Jul 2024
- EcoSta 2023, Tokyo, Japan Aug 2023

Reviewer, for many journals including the following:

- Annals of Applied Statistics, Biometrics, Electronic Journal of Statistics, Journal of Multivariate Analysis
- Journal of the American Statistical Association, Journal of the Royal Statistical Society
- Journal of Sports Analytics, Sankhya A and Sankhya B, Statistics and Probability Letters

SKILLS

Technical strength:

- Proficient: R, Python, $\text{\LaTeX}$, Microsoft Office, Google Suite
- Working knowledge: MATLAB, SQL, C, C++

Languages:

- Bilingual: English, Bengali; Intermediate: Hindi; Basic: Portuguese.